

Qianqian (Bella) He

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SUMMARY

University of Chicago M.S. Computer Science candidate graduating December 2026 with 5+ years building production AI systems, agentic workflows and scalable data platforms across Alibaba, L'Oréal, Trina Solar and research labs. Built systems spanning 10M+ multimodal and 2M+ time-series data with \$8M+ annual impact. Research focuses on long-horizon agent, evaluation, and ML system.

SKILLS

- AI/ML: LLMs, Agentic AI, RAG, RL, Multimodal ML, ML Evaluation, Model Inference, NLP, CV, PyTorch, TensorFlow
- Engineering: Python, C++, Java, SQL, JavaScript, Go, R, CUDA, Distributed System, API, Docker, Kubernetes, AI Infrastructure
- Data Science: Governance, Modeling, Pipelines, Database, Spark, Statistics, Experiment Design, Optimization, Forecasting

EDUCATION

The University of Chicago - M.S. in Computer Science, Chicago, IL

Sept 2025 - Dec 2026

Relevant Course: Machine Learning, NLP, Computer Vision, Distributed System, Parallel Computing, Advanced C++, Optimization

PROFESSIONAL EXPERIENCE

UChicago Data Science Institute - Research Intern, Prof. Chenhao Tan, Chicago, IL

June 2026 - Present

- Built multi-agent data and evaluation infrastructure supporting 1,126 decision points from 9 runs with canonical event schemas, provenance metadata, trajectory visualization, and human-in-the-loop verification.
- Developed LLM tool use and predictive modeling workflows for 40 phase-action and 107 semantic classes, achieving 0.294 Recall@1 0.294 vs. 0.125 random on future trajectory-state prediction.
- Extended trajectory prediction into dynamic-agent recovery across 16 disruption episodes and a real-agent case, achieving 93.8% final-state correctness and 100% rollback correctness.

Trina Solar Co., Ltd. - Data Scientist, Shanghai, CN

Jan 2024 – Aug 2025

- Architected an agentic supply chain optimization system using LLMs, RAG, forecasting, and workflow orchestration across pricing, supplier, logistics, and market intelligence data, achieving \$6.65M in annual cost savings.
- Built a production energy management platform integrating CNN-LSTM forecasting, CPLEX optimization, reinforcement learning, real-time data ingestion, and APIs across 2M+ time-series datasets, reducing annual operational costs by \$1.53M.
- Built enterprise data governance and analytics foundations, including data catalogs, standards, models and pipelines across supply chain and energy operations to support consistent analysis and decision-making.

L'Oréal S.A. - Data Analytics & Governance Manager, Shanghai, CN

Nov 2022 - Dec 2023

- Optimized a transformer-based multimodal recommendation system across 10M+ structured and unstructured datasets, improving recommendation relevance and user satisfaction by 13%.
- Developed a GenAI analytics platform combining ETL automation, natural language querying, predictive analytics, NLP-based insight generation, and dashboards, improving operational efficiency by 21.4%.
- Built multi-channel online/offline data pipelines and a governed analytics platform with standardized data catalogs, domain data models, metadata management, data quality controls, and dashboard analytics.

AISHU Technology Corp. - Product Manager, Shanghai, CN

Aug 2021 – Oct 2022

- Led development of an AIOps platform using Java and Rust for anomaly detection, failure prediction, infrastructure monitoring, and root-cause analysis across distributed systems, improving operational efficiency by 30%.
- Designed telemetry processing pipelines and real-time monitoring services supporting servers, networks, and application infrastructure at scale.
- Implemented MLOps and backend orchestration workflows for event tracking, observability, and operational analytics, improving system scalability, reliability, and incident response.

Alibaba Group Holding Ltd. - Product Manager, Hangzhou, CN

May 2020 - July 2021

- Managed development and lifecycle of the Alibaba Tianchi Big Data & AI Platform, using user analytics, rapid prototyping, open-source datasets, and AI model deployment to improve product adoption.
- Collaborated with engineering teams on platform architecture, testing, CI/CD, and developer workflows, expanding platform capabilities and improving development efficiency.
- Developed a medical image classification model using ResNet34, FPN and BBN mixup in Python and Pytorch, improving macro F1 by 20.7% for model benchmarking and evaluation.

RESEARCH EXPERIENCE & PUBLICATIONS

QIAO-Bench – Evaluating Predictive World Models for Multi-Agent System

WM@Booth Workshop 2026

- Built a leakage-resistant benchmark for future state prediction in multi-agent workflows, curating 1,126 decision points from 9 decision-bearing runs across 40 phase-action classes and 107 semantic classes with grouped run-level evaluation splits.
- Evaluated random, TF-IDF, ridge regression, and JEPA-inspired predictors across state-only, action-only, and state-action inputs; gated state-action JEPA reached Recall@1 0.294 vs. 0.125 random, while action-only JEPA achieved ~3.5x random Recall@10.

RecoverFlow – Evaluating Consistency and Recovery in Long-Horizon Agents

EMNLP Workshop 2026

- Developed a matched-counterfactual evaluation and recovery framework for long-horizon tool-using agents under evolving user goals and environment states, using goal-versioned trajectories, recovery dispositions, and dependency-aware revised planning.
- Evaluated on 16 disruption episodes plus a real-agent matched pair, achieving 0.968 dependency F1, 0.964 disposition macro-F1, 93.8% final-state correctness and 100% rollback correctness, outperforming full planning (68.8%) and restart (75.0%).

Explainable AI for Reliable Energy Consumption Forecasting – A SHAP-based Approach

EIECT 2024

- Built an end-to-end time series forecasting pipeline using 34,895 train and 1,776 test electricity data, engineered 22 lag, rolling, and temporal feature and compared 5 ML models including XGBoost, LightGBM, random forest, decision tree, and SVM.
- Achieved best performance with XGBoost (MAE 0.0297, MSE 0.00169) and applied SHAP to quantify model-specific feature contributions and improve prediction interpretability and transparency.

NTU Business AI Lab - Research Intern, Dr. Teoh Teik Toe

Jan 2023 - Mar 2023

- Developed a CNN for 3-class breast-ultrasound classification using 1,578 500x500 images from 600 patients, with 1,103 train/157 valid/318 test images, convolution/max-pooling layers, dropout, and image augmentation.
- Evaluated augmentation strategies, with random rotation and flipping achieving 92.1% test accuracy, versus 91.4% baseline and 89.3% with resize/rescale, demonstrating improved generalization and reduced overfitting.

SELECTED PROJECTS

Agent Security Red-Teaming & Evaluation: Multi-Step Tool Attacks by Kaggle | Python, Agent Eval, Tool-Use Security

- Built an algorithmic red-teaming system for tool-using agents, discovering replayable multi-step attack trajectories across 4 security failure classes and 2 target model families.
- Modeled execution as structured state-action-observation-tool trajectories, capturing prompts, memory, tool calls, file/API operations, and policy outcomes for reproducible failure localization and causal analysis.
- Extended safety evaluation toward predictive failure detection and recovery, identifying unsafe-transition boundaries, attributing failures to preceding actions, and determining when agents should continue, verify, replan, or reverse prior actions.

ML Systems & GPU Inference Optimization | C++, CUDA, PyTorch, ONNX, Parallel Computing

- Built an end-to-end neural-network inference and evaluation pipeline with ONNX graph generation and validation, tensor encoding, batched processing, and parallel data loading for reproducible ML workloads.
- Profiled 4 major performance bottleneck classes – memory layout, batching, kernel execution, and CPU/GPU transfers and applied CUDA-aware optimization patterns across the inference path.
- Developed experiment configuration, model-validation checks, and reproducible run artifacts to improve evaluation repeatability, systems debugging, and reliability for large-scale model experiments.

Agentic Portfolio Management and Predictive Market Intelligence | Java, Python, OOP, LangChain, ML Forecasting

- Architected an event-driven financial platform across 4 service domains—market data, transaction, recommendations, and portfolio management—using ActiveMQ and Apache Camel for asynchronous data and decision workflows.
- Built a market-forecasting and uncertainty layer to generate predictive signals from historical market data, evaluating forecasts with statistical and calibration metrics and incorporating model confidence into portfolio risk and rebalancing decisions.
- Developed LLM-based agentic workflows and tool calling that combine forecasts, market conditions, portfolio state, and investor risk profiles to automate investment analysis, risk evaluation, recommendation, and dynamic rebalancing.

Distributed Energy Simulation & Agent Control Platform | Python, FastAPI, Docker, Distributed Systems

- Developed a distributed energy simulation platform with FastAPI and Docker, exposing authenticated REST APIs and encrypted inter-service communication for state monitoring, scheduling, and energy-control workflows.
- Implemented fault-tolerant coordination through message relaying, automated service re-registration, and hybrid in-memory/relational state management, preserving consistency under distributed service failures.
- Extend the simulator toward an agent RL environment with explicit state, action, transition, and evaluation interfaces, enabling trajectory logging and future work on adaptive control, failure recovery, and reinforcement-learning policies.